MIL-N-60209(MU)
4 March 1965
SUPERSEDING
FA(FC)-PD-104
27 April 1961

MILITARY SPECIFICATION

NETWORK A: 10526472

1. SCOPE

1.1 This specification covers one type of network circuit board, part of the Gun Direction Computer: M18. The network consists of output drivers, input drivers, logic driver, neon driver No. 1, neon driver No. II, matrix decoders, back contact clamps, "And" gate, and ground clamps (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction: M18 Manufacturing processes and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWING

U.S. Army Munitions Command

F10526472	Network A

PACKAGING DATA SHEET

M14002	Data Sheet, Master Packaging

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:18
Check the source to verify that this is the current version before use.

FSC 1220

(Copies of specifications, standards, drawings, and packaging data sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The network shall be manufactured in accordance with Drawing F10526472 and drawings pertaining thereto.

3.2 General specifications.- The network shall meet the following requirements of MIL-F-13926, and MIL-C-46357.

3.2.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Workmanship

3.2.2 MIL-C-46357

a. Etched circuit assembly
b. Marking
c. Handling

3.3 Environmental service conditions.

3.3.1 Vibration.- The network shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion). The frequency shall be varied uniformly from 10 to 55 cycles per second (cps) and return to 10 cycles. This cycle shall be traversed in 1 minute $\pm$ 15 seconds. The motion shall be applied to the network for a period of 10 minutes in each direction of the 3 mutually perpendicular axes (total of 30 minutes), relative to its normal operational position. Subsequent to vibration, the network shall not be damaged or adversely affected and shall meet the requirements of 3.4.

3.3.2 Shock.- The network shall withstand 3 shock impulses in the vertical plane, relative to its normal mounting position. The acceleration of the shock impulses shall be 5 gravity units (G's). Pulse duration of each impulse shall be 11 $\pm$ 1 milliseconds. Subsequent to shock the item shall not have been damaged or adversely affected and shall meet the requirements of 3.4.

3.3.3 Temperatures.

3.3.3.1 Operating.- The network shall meet the requirements of 3.4.1.3, 3.4.2.5, 3.4.3.4, 3.4.4.5, 3.4.5.3 and 3.4.5.4 while exposed to temperatures of -40°F and +135°F.

3.3.3.2 Storage.- The network shall meet the requirements of 3.4 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80°F and +160°F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:18Z
Check the source to verify that this is the current version before use.

3.3.4 Humidity.

3.3.4.1 Operating.- The network shall meet the requirements of 3.4.1.3, 3.4.2.5, 3.4.3.4, 3.4.4.5, 3.4.5.3 and 3.4.5.4 while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of 135°F.

3.3.4.2 Non-operating.- The network shall meet the requirements of 3.4 after exposure to (and when thoroughly dried) an atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.3.5 Fungus resistance.- The network shall show no evidence of damage and shall meet the requirements of 3.4, after being inoculated with the composite spore suspension specified in Specification MIL-F-13926 and incubated at an ambient temperature of between 77°F and 84°F with a RH of 97 ± 2%.

3.3.6 Salt spray.- The network shall show no evidence of physical damage and meet the requirements of 3.4 immediately after exposure to a fine mist of a 5% salt solution concentration. The solution shall have a specific gravity of from 1.0268 to 1.0413 and a pH between 6.5 and 7.2 at a temperature between 93° and 97°F.

3.4 Performance.- Unless otherwise specified the network shall meet the performance requirements of this specification at ambient temperatures between 60° and 90°F; at a barometric pressure of 28 to 32 inches of mercury; and a RH up to 90%.

3.4.1 Output driver.

3.4.1.1 True input.- Static true input to output driver.

(a) The input transistor collector to emitter voltage shall be less than 0.3 volt.

(b) The output transistor collector to emitter voltage shall be less than 0.5 volt.

3.4.1.2 False input.- Static false input to Output Driver.

(a) The input transistor leakage shall be less than 15 micro-amperes.

(b) The output transistor leakage shall be less than 25 micro-amperes.

(c) The clamp diode across the output transistor shall have a forward voltage drop of 0.5 ± 0.4 volt.

3.4.1.3 Dynamic input.- The output wave form shall possess the characteristics of Figure 1. The delay time measured from the time when the input is at minus 1.5 volts (going toward minus 6 volts), to the time when the output reaches minus 2 volts shall be less than 1 microsecond. The delay time measured from the time when the input Figure 2 is at minus 1.5 volts (going toward 0 volts), to the time when the output reaches minus 1 volt shall be less than 2 microseconds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:18Z
Check the source to verify that this is the current version before use.

3.4.2 Input driver.

3.4.2.1 Clamp diode.- The clamp diode connected to the emitter of the input transistor shall have a forward drop of less than 0.5 volt at 5 ± 0.5 milliamperes and its reverse leakage shall be less than 30 micro-amperes with 9.5 volts reverse voltage.

3.4.2.2 Ground clamp.- The ground clamp diode at the output of the input driver shall have a forward drop of less than 0.8 volt at 30 ± 5 milliamperes.

3.4.2.3 Input transistor.- The input transistor collector to emitter drop shall be less than 0.5 volt when it is forward biased. The collector leakage current shall be less than 25 micro-amperes when it is reversed biased.

3.4.2.4 Minus 6 volt clamp diode.- The minus 6 volt clamp diode at the driver collector shall have a forward drop of less than 0.7 volt at 0.5 ± 0.1 milliamperes.

3.4.2.5 Dynamic input.- The output wave form shall possess the characteristics of Figure 3 when an input wave form of Figure 4 is applied. The delay time measured when the input is at minus 1.5 volts (going toward minus 6 volts), to the time when the output reaches minus 1.5 volts shall be less than 2 microseconds. The delay time measured from the time when the input is at minus 1.5 volts (going toward ground) to the time when the output is at minus 1.5 volts, shall be less than 1 microsecond.

3.4.3 Neon driver No. I.

3.4.3.1 Input diodes.- The input diodes shall have a forward voltage drop of less than 0.8 volt at 3 ± 1 milliamperes and a reverse leakage of less than 60 microamperes.

3.4.3.2 Output transistor.- The voltage level at TP10 with the load of Figure 5 shall be plus 115 ± 15 volts with a false input and shall be plus 23 ± 15 volts with a true input.

3.4.3.3 Input transistor.- The base-emitter junction shall have a forward voltage drop of less than 1.0 volt.

3.4.3.4 Dynamic input.- The output waveform shall possess the characteristics of Figure 6, when supplied with an input waveform of Figure 7. The delay time measured from the time the input is at minus 3 (going toward minus 6 volts) to the time when the output reaches plus 90 volts shall be less than 10 milliseconds. The delay time measured from the time when the input is at minus 3 volts (going toward 0 volts), to the time when the output reaches plus 90 volts shall be less than 10 milliseconds.

3.4.4 Neon driver No. II.

3.4.4.1 Input diodes.- The input diodes shall have a forward voltage drop of less than 0.5 volt at 0.5 ± 0.1 milliampere and a reverse leakage of less than 60 microamperes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:18Z
Check the source to verify that this is the current version before use.

3.4.4.2 Threshold diode.- The threshold diode shall have a forward voltage drop of less than 0.5 volt at 2.5 ± 0.5 milliamperes and a reverse leakage of less than 60 microamperes.

3.4.4.3 Output transistor.- The output transistor leakage shall be less than 10 microamperes.

3.4.4.4 Input transistor.- The base to emitter junction, shall have a forward voltage drop of less than 1.0 volt.

3.4.4.5 Dynamic input.- The output waveform shall possess the characteristics of Figure 6, when supplied with an input waveform of Figure 7. The delay time measured from the time the input is at minus 3 volts (going toward minus 6 volts), to the time when the output reaches plus 90 volts, shall be less than 10 milliseconds. The delay time measured from the time the input is at minus 3 volts (going toward 0 volts) to the time when the output reaches plus 90 volts, shall be less than 10 milliseconds.

3.4.5 Logic driver.

3.4.5.1 True-state static condition driver unloaded.- With a true input (Figure 8), the following condition shall exist with the driver:

(a) The saturation voltage of input switch shall not exceed 0.3 volt.

(b) The driver transistor leakage current shall not exceed 15 microamperes.

(c) The driver transistor base to emitter voltage shall be greater than 0.2 volt, but less than 1.5 volts.

3.4.5.2 False-state static condition, driver loaded.- With a false signal applied to the input gate (Figure 8), the following condition shall exist with the driver.

(a) The zener voltage shall be 6.3 ± .6 volts.

(b) The saturation voltage of the driver transistor shall not exceed 0.3 volt.

(c) The emitter to base voltage of the driver transistor shall not exceed 0.7 volt.

3.4.5.3 Dynamic condition driver loaded.- With a symmetrical square wave varying from zero to minus 6 volts applied to it, the input gate (Figure 8), the output of the driver shall be a square wave swinging between the levels of zero and minus 6 volts.

3.4.5.4 Delay time comparison of input and output waveform.- The delay time measured from the time the square wave applied to the input gate reaches minus 1.5 volts (going toward minus 6 volts) until the output reaches 90 percent of its total excursion (going toward minus 6 volts) shall be equal to or less than 150 millimicroseconds. The delay time measured from the time the square wave applied to the input gate reaches minus 1.5 volts (going toward zero volts) until the output reaches 90 percent of its total excursion (going toward zero volts) shall be equal to or less than 150 millimicroseconds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:18Z
Check the source to verify that this is the current version before use.

3.4.6 Static test for matrix decoder, back contact clamps, ground clamps, "And" gates, gate and diode.

3.4.6.1 1N99 diodes.- The forward voltage drop shall be less than 1 volt at 3 ± 0.5 milliamperes and its reverse of leakage shall be less than 40 microamperes at 18 ± 4 volts.

3.4.6.2 CTP632 diodes.- The forward voltage drop shall be less than 0.65 volt and its reverse leakage shall be less than 10 microamperes at 15 ± 3 volts.

3.5 Wiring.- Wiring and connections shall be in accordance with electrical schematic, Drawing F10526475.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of Product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examinations and tests.- Except where otherwise specified herein, components and sub-assemblies shall be inspected in accordance with the "Inspection provisions for components and sub-assemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables I, II and III and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.4 Functional tests.- The requirements and tests in Table I shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:18Z
Check the source to verify that this is the current version before use.

TABLE I
(100 percent inspection)

Characteristics	Requirement	Test Procedures
101. True input-static, output driver (O.D.)	3.4.1.1	4.5.2.2
102. False input-static (O.D.)	3.4.1.2	4.5.2.2.4
103. Dynamic input (O.D.)	3.4.1.3	4.5.2.2.5
104. Clamp diode, input driver (I.D.)	3.4.2.1	4.5.2.3.1
105. Ground clamp, (I.D.)	3.4.2.2	4.5.2.3.2.1
106. Input transistor, (I.D.)	3.4.2.3	4.5.2.3.3, 4.5.2.3.3.3, 4.5.2.3.3.4
107. Minus 6 volt clamp diode, (I.D.)	3.4.2.4	4.5.2.3.3.4
108. Dynamic input, (I.D.)	3.4.2.5	4.5.2.3.3.5
109. Input diodes, neon driver No. 1 (N.D. #1)	3.4.3.1	4.5.2.4.1
110. Output transistor, (N.D. #1)	3.4.3.2	4.5.2.4.2
111. Input transistor, (N.D. #1)	3.4.3.3	4.5.2.4.2.4
112. Dynamic input (N.D. #1)	3.4.3.4	4.5.2.4.2.5
113. Input diodes, neon driver No. 11 (N.D. #11)	3.4.4.1	4.5.2.5.1
114. Threshold diodes, (N.D. #11)	3.4.4.2	4.5.2.5.2
115. Output transistor, (N.D. #11)	3.4.4.3	4.5.2.5.3
116. Input transistor, (N.D. #11)	3.4.4.4	4.5.2.5.3.4
117. Dynamic input, (N.D. #11)	3.4.4.5	4.5.2.5.3.5
118. True-state static condition, driver unloaded, logic driver (L.D.)	3.4.5.1	4.5.2.5.4.3
119. False-state static condition, driver loaded, (L.D.)	3.4.5.2	4.5.2.5.4.4
120. Dynamic condition, driver loaded (L.D.)	3.4.5.3	4.5.2.5.4.5
121. Delay time comparison of input	3.4.5.4	4.5.2.5.4.5.3
122. Static test for matrix decoder, back contact clamps, ground clamps, "And" gates, gate and diode (1N99 & CTP632)	3.4.6.1	4.5.2.5.5

4.3 Standard sampling.

4.3.1 General.- One network shall be selected at random as a standard sample from each 100 produced. The standard sample shall meet the requirements and tests in Table II.

TABLE II

Characteristics	Requirement	Test Procedures
301. Fabrication	3.1	Applicable drawings - visual
302. General specifications	3.2	MIL-F-13926 and MIL-C-46357 - visual
303. Wiring	3.5	4.5.3 - visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:18Z
Check the source to verify that this is the current version before use.

4.3.2 Environmental.- Three units shall be selected at random by a Government representative as standard samples from each 50 produced or from each month's production, whichever occurs first. The standard samples shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table III.

TABLE III

Characteristics	Requirement	Test Procedures
303. Vibration	3.3.1	4.5.1.1
304. Shock	3.3.2	4.5.1.2
305. Humidity-Operating	3.3.4.1, 3.4.1.3, 3.4.2.5, 3.4.3.4, 3.4.4.5, 3.4.5.3, 3.4.5.4	4.5.1.3.2, 4.5.2.2.6, 4.5.2.3.3.6, 4.5.2.4.2.6, 4.5.2.5.3.6, 4.5.2.5.4.6
306. Humidity-Non-operating	3.3.4.2	4.5.1.3.1
307. Salt spray	3.3.6	4.5.1.4
308. Operating temperature	3.3.3.1, 3.4.1.3, 3.4.2.5, 3.4.3.4 3.4.4.5, 3.4.5.3, 3.4.5.4	4.5.1.4, 4.5.2.2.6, 4.5.2.3.3.6, 4.5.2.4.2.6, 4.5.2.5.3.6, 4.5.2.5.4.6
309. Storage temperature	3.3.3.2	4.5.1.5

4.3.3 Failure of sample.- Should any one item of a standard sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by theGovernment until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.2.3).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Test equipment.- In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:18Z
Check the source to verify that this is the current version before use.

Item Number	Description
1	Oscilloscope - A dual trace oscilloscope with a rise time of less than 50 millimicroseconds and an input impedance of 1 meg. ohm or greater and a input capacitance of less than 30 micromicrofarad.
2	Voltage measuring device - A + 5% D.C. voltage measuring device with a 10 millivolt to 300 volt range. The input impedance shall be greater than 10 meg ohms.
3	Pulse generator - A generator of pulses capable of square waves of 250 ± 25 K.c. with a rise and fall time of less than 150 millimicroseconds and an output impedance of less than 150 ohms.
4	Power supply - Minus 1.5 volts ± 2 percent with a current requirement of minus 150 milliamperes. The peak-to-peak ripple and noise shall be less than 100 m.v.. The output impedance shall be less than 2 ohms from d.c. to 500 k.c.
5	Power supply - Minus 3 volts ± 2 percent, current requirement of minus 20 milliamperes. The peak-to-peak ripple and noise shall be less than 100 m.v. The output impedance shall be less than 2 ohms from d.c. to 500 k.c.
6	Power supply - Minus 6 volts ± 2 percent, current requirement of 200 milliamperes. The peak-to-peak and noise shall be less than 400 m.v. and the output impedance shall be less than 2 ohms from d.c. to 500 k.c.
7	Power supply - Minus 18 volts ± 2 percent, current requirement of 50 milliamperes. The peak-to-peak ripple and noise shall be less than 600 m.v. and the output impedance shall be less than 2 ohms from d.c. to 500 k.c.
8	Power supply - Plus 3 volts ± 2 percent, current requirement of 50 milliamperes. The peak-to-peak ripple and noise shall be less than 100 m.v., and the output impedance shall be less than 2 ohms from d.c. to 500 k.c.
9	Power supply - Plus 6 volts ± 2 percent, current requirement of 50 milliamperes. The peak-to-peak ripple and noise shall be less than 100 m.v., and the output impedance shall be less than 2 ohms from d.c. to 500 k.c.
10	Power supply - Plus 15 volts ± 2 percent, current requirement of 50 milliamperes. The peak-to-peak ripple and noise shall be less than 100 m.v., and the output impedance shall be less than 2 ohms from d.c. to 500 k.c.
11	Power supply - Plus 155 volts ± 2 percent, current requirement of 20 milliamperes. The peak-to-peak ripple and noise shall be less than 400 m.v. and the output impedance shall be less than 10 ohms from d.c. to 500 k.c.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:18Z
Check the source to verify that this is the current version before use.

Item Number	Description
12	Power supply - Plus 66 volts minus 12, plus 9 percent, current requirement of 20 milliamperes. The peak-to-peak ripple and noise shall be less than 200 m.v., and the output impedance shall be less than 10 ohms from d.c. to 500 k.c.
13	Power supply - Variable, from zero to one hundred volts, current requirement of 100 milliamperes. The peak-to-peak ripple and noise shall be less than 500 m.v. and the output impedance shall be less than 10 ohms from d.c. to 500 k.c.

4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Vibration.- The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1 with the network rigidly mounted in its normal operating position. Subsequent to the vibration test, the unit shall show no damage or defect resulting from the test when examined by visual and tactile means. The unit shall then be inspected as specified in 4.5.1.2 prior to determining compliance with requirements specified under 3.4.

4.5.1.2 Shock.- This test shall be conducted in accordance with the conditions of 3.3.2 and Procedure II of the shock test defined in MIL-F-13926 except that impact shocks shall not exceed 5 G's. The network shall be subsequently inspected for physical damage or defects in compliance with 3.3.2. The unit shall then be inspected as specified in 4.5.1.3 prior to determining compliance with the requirements of 3.4 as required by 3.3.2.

4.5.1.3 Humidity.

4.5.1.3.1 Non-operating.- This test shall be conducted in accordance with the conditions in 3.3.4.2 and the humidity test defined in Procedure I of MIL-F-13926 except that the 4-1/2 hour period shall be reduced to 2 hours, the 6 hour period reduced to 2 hours, and the total test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dried and then subjected to the test in 4.5.1.3.2 prior to meeting the specified requirements of 3.4 as required by 3.3.4.2.

4.5.1.3.2 Operating.- This test shall be conducted in accordance with the conditions in 3.3.4.1 and the humidity test defined in Procedure III of MIL-F-13926 except that the humidity shall be maintained at 97 $\pm$ 5 percent for the 24 hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test the unit shall be operated in accordance with 4.5.2.2.6, 4.5.2.3.3.6, 4.5.2.4.2.6, 4.5.2.5.3.6 and 4.5.2.5.4.6 to determine compliance with 3.3.4.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:18Z
Check the source to verify that this is the current version before use.

4.5.1.4 Salt spray.- The network shall be subjected to the conditions in 3.3.6 and the salt spray test defined in MIL-STD-202, Condition B. Immediately after exposure, the unit shall be inspected for any physical damage, then tested in accordance with 4.5.2.2 to 4.5.2.5.5.3 inclusive for compliance with 3.3.6.

4.5.1.5 Temperature (Operating and storage).- After being subjected to all environmental tests specified above, the network shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperatures specified in 3.3.3 shall apply. While at the operating temperatures of 3.3.3.1, the unit shall be tested in accordance with 4.5.2.2.6, 4.5.2.3.3.6, 4.5.2.4.2.6, 4.5.2.5.3.6 and 4.5.2.5.4.6 to determine compliance with the requirements of 3.4.1.3, 3.4.2.5, 3.4.3.4, 3.4.4.5 and 3.4.5.4 as required by 3.3.3.1. The temperature shall then be returned to standard temperature (60° to 90°F) and the unit shall be tested in accordance with 4.5.2.2 to 4.5.2.5.5.3 inclusive to determine compliance with 3.3.3.2.

4.5.1.6 Fungus.- To determine compliance with 3.3.5, one of the first 5 networks produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

4.5.2 Performance.- The subsequent tests shall be performed in accordance with the conditions specified in 3.4, except where otherwise specified.

4.5.2.1 Voltage polarity convention.- Whenever the potential difference existing between two points is measured, one of the two points shall be labeled the "From" point and the other shall be labeled the "To" point. The measured potential difference shall be a positive quantity if the potential at the "From" point is more positive than that at the "To" point.

4.5.2.2 Output driver.- To determine compliance with 3.4.1, the following conditions shall be applied for the conduct of tests herein.

4.5.2.2.1 Input gates and loads.- Connect to the pins listed in Table IV an input gate (Figure 9) to the inputs and a load (Figure 12) to the outputs.

TABLE IV

INPUT GATES, TO PINS	LOADS, TO PINS
32B	34B
36B	37B
39B	40B
43B	44B
46B	48B

4.5.2.2.2 Voltage supply.- Connect to pins listed in Table V, the corresponding voltage supply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:18Z
Check the source to verify that this is the current version before use.

TABLE V

PINS	VOLTAGE	ITEM NO
2A	-1.5 V	4
54B	-3.0 V	5
52A	-6.0 V	6
5B	-18 V	7
2B	+3 V	8
30A	+6 V	9
25B	+15 V	10
65B	+155 V	11
60B	+66 V	12
23B	Ground	
24B	Ground	

4.5.2.2.3 True input.- Connect the minus 6 volt supply voltage to the inputs of each input gate. Measure the voltage from those points listed in Column 1 to that of Column 2. The voltage shall be as in Column 3 of Table VI to conform with 3.4.1.1 (a) and (b).

TABLE VI

COLUMN 1, FROM PIN	COLUMN 2, TO PIN	COLUMN 3, VOLTAGE
2A	31B	$\leq$ +0.30
2A	35B	$\leq$ +0.30
2A	38B	$\leq$ +0.30
2A	41B	$\leq$ +0.30
2A	45B	$\leq$ +0.30
TP6	52A	$\leq$ +0.50
TP7	52A	$\leq$ +0.50
TP8	52A	$\leq$ +0.50
TP9	52A	$\leq$ +0.50
50B	52A	$\leq$ +0.50

4.5.2.2.4 False input.- Remove the minus 6 volt supply voltage from the input gates inputs and apply ground to these inputs. Measure the voltage from those points listed in column 1 to that of column 2 and the voltage shall be as in column 3 of Table VII to conform with 3.4.1.2(a), (b) and (c).

TABLE VII

COLUMN 1, FROM PIN	COLUMN 2, TO PIN	COLUMN 3, VOLTAGE
31B	TP1	$\leq$ +0.012
35B	TP2	$\leq$ +0.012
38B	TP3	$\leq$ +0.012
41B	TP4	$\leq$ +0.012
45B	TP5	$\leq$ +0.012
52A	TP1	+0.50 $\pm$ 0.40
52A	TP2	+0.50 $\pm$ 0.40
52A	TP3	+0.50 $\pm$ 0.40
52A	TP4	+0.50 $\pm$ 0.40

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:18:
Check the source to verify that this is the current version before use.

COLUMN 1, FROM PIN	COLUMN 2, TO PIN	COLUMN 3, VOLTAGE
52A	TP5	+0.50 ± 0.40
23B	34B	≤+0.150
23B	37B	≤+0.150
23B	40B	≤+0.150
23B	44B	≤+0.150
23B	48B	≤+0.150

4.5.2.2.5 Dynamic input.- Remove the ground from the inputs of each input gate and apply a 5Kc ± 10 percent square wave varying from 0 ± 0.5 volts to minus 6 + 1 volts. The 10 to 90 percent rise and fall time shall be less than 0.5 microseconds as shown in Figure 2. The output wave form shall be as shown in Figure 1. The delay time measured from the time the input is at minus 1.5 volts (going toward minus 6 volts), to the time when the output reaches minus 2 volts, shall be less than 1 microseconds. The delay time measured from the time the input is at minus 1.5 volts (going to 0 volts), to the time when the output reaches minus 1 volts, shall be less than 2 microseconds. Measure the above mentioned output wave forms at pins 34B, 37B, 40B, 44B and 48B to determine compliance with 3.4.1.3.

4.5.2.2.6 Circuit output.- The output driver shall be subjected to the test of paragraph 4.5.2.2.5 when thermally stabilized at ambient temperatures of minus 40 degrees and plus 135 degrees Fahrenheit for compliance with 3.3.3.1.

4.5.2.3 Input driver.

4.5.2.3.1 Clamp diode.

4.5.2.3.1.1 Forward voltage.- Disconnect all D.C. Power Supplies. Connect a minus 1.5 volts and ground to pins 52A and 2A respectively. The voltage measured from pin 2A to following test points shall be less than +0.5 volt: Test points 11, 12, 13, 14 and 15 to determine compliance with 3.4.2.1. Remove the connections to pins 52A and 2A.

4.5.2.3.1.2 Reverse leakage.- Connect the minus 1.5 volt power supply and ground pin 52A. The voltage measured from pin 52A to the following test points shall be less than plus 0.025 volt; test points 11, 12, 13, 14 and 15 to determine compliance with 3.4.2.1. Remove all test leads after test.

4.5.2.3.2 Ground clamp.

4.5.2.3.2.1 Forward voltage.- Connect the minus 6 volts in series with a 1K ± 5% resistor to pin 23B and ground pin 52A. The voltage measured from pin 23B to the following pins shall be minus 0.35 ± 0.30 volt; pins 7B, 11B, 15B, 19B and 26B to determine compliance with 3.4.2.2. Remove the connections from pins 23B and 52A.

4.5.2.3.3 Input transistor.

4.5.2.3.3.1 Load.- Connect to each of pins 7B, 11B, 15B, 19B and 26B a 3.3K ohms, 1/2 watt, ± 5 percent load resistor from the minus 18 volt supply voltage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:18Z
Check the source to verify that this is the current version before use.

4.5.2.3.3.2 Power supply.- Connect power to the appropriate pins as shown in Table V.

4.5.2.3.3.3 False input.- Ground pins 8B, 12B, 17B, 20B and 28B. The voltage measured from those points listed in column 1 to column 2 shall be as specified in column 3 of Table VIII to determine compliance with 3.4.2.3.

TABLE VIII

COLUMN 1, FROM PIN	COLUMN 2, TO PIN	COLUMN 3, VOLTAGE
10B	TP11	$\leq$ +0.50
13B	TP12	$\leq$ +0.50
18B	TP13	$\leq$ +0.50
22B	TP14	$\leq$ +0.50
30B	TP15	$\leq$ +0.50
7B	23B	$\leq$ +0.80
11B	23B	$\leq$ +0.80
15B	23B	$\leq$ +0.80
19B	23B	$\leq$ +0.80
26B	23B	$\leq$ +0.80

4.5.2.3.3.4 True input.- Remove ground from pins 8B, 12B, 17B, 20B and 28B and connect a minus 6 volt supply voltage to these pins. The voltage measured from those points listed in column 1 to column 2 shall be as listed in column 3 of Table IX to determine compliance with 3.4.2.3 and 3.4.2.4.

TABLE IX

COLUMN 1, FROM PIN	COLUMN 2, TO PIN	COLUMN 3, VOLTAGE
30A	10B	$\leq$ +0.20
30A	13B	$\leq$ +0.20
30A	18B	$\leq$ +0.20
30A	22B	$\leq$ +0.20
30A	30B	$\leq$ +0.20
52A	7B	$\geq$ +0.70
52A	11B	$\geq$ +0.70
52A	15B	$\geq$ +0.70
52A	19B	$\geq$ +0.70
52A	26B	$\geq$ +0.70

4.5.2.3.3.5 Dynamic input.- Repeat paragraph 4.5.2.3.3.1 and 4.5.2.3.3.2, and apply to pins 8B, 12B, 17B, 20B and 28B, a 5 KC $\pm$ 10 percent square wave varying from 0 $\pm$ 1 volt to minus 4 $\pm$ 2 volts. The 10 to 90 percent rise and fall time shall be less than 0.5 microsecond as shown in Figure 4. The output wave forms at pins 7B, 11B, 15B, 19B and 26B shall be as shown in Figure 3. The output wave form shall have a delay of less than 2 microseconds measured from the time when the input is at minus 1.5 volts (going toward minus 6 volts) to the time when the output reaches minus 1.5 volts. The delay time measured from the time when the input is at minus 1.5 volts (going toward 0 volts), to the time when the output is minus 1.5 volts shall be less than 1 microsecond to conform with 3.4.2.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:18Z
Check the source to verify that this is the current version before use.

4.5.2.3.3.6 Circuit output.- The input driver shall be subjected to the test of paragraph 4.5.2.3.3.5 when thermally stabilized at ambient temperatures of minus 40 degrees to plus 135 degrees Fahrenheit for compliance with 3.3.3.1.

4.5.2.4 Neon driver No. 1.

4.5.2.4.1 Input diodes.

4.5.2.4.1.1 Forward voltage drop.- Connect the minus 18 volt supply voltage to pin 5B. The voltage measured from pins 61B, 63B, and 64A, when each is grounded one at a time, to pin 62B shall be less than +0.8 volt to conform with 3.4.3.1. Remove all test leads upon completion of test.

4.5.2.4.1.2 Reverse leakage.- Ground pin 5B. The voltage measured from pin 5B to pin 62B, when a minus 18 volt supply voltage is applied one at a time to pins 61B, 63B and 64A shall be less than plus 0.35 volt, to conform with 3.4.3.1. Remove all test leads upon completion of test.

4.5.2.4.2 Output transistor.

4.5.2.4.2.1 Load.- Connect to pin 65A the load shown in Figure 5.

4.5.2.4.2.2 Power supplies.- Connect power to the appropriate pins as shown in Table V.

4.5.2.4.2.3 False input.- Ground pin 61B and measure the voltage from TP10 to ground. The voltage shall be plus 112 $\pm$ 15 volt to conform with 3.4.3.2.

4.5.2.4.2.4 True input.- Remove the ground from pin 61B and apply a minus 6 volt supply voltage to it. The voltage measured from the points listed in column 1 to column 2 shall be as listed in column 3 of Table X to conform with 3.4.3.2 and 3.4.3.3. Remove connections to pin 61B upon completion of test.

TABLE X

COLUMN 1, FROM PIN	COLUMN 2, TO PIN	COLUMN 3, VOLTAGE
54B	62B	$<$+1.0
TP10	23B	+23 $\pm$ 15
65A	23B	+75 $\pm$ 9

4.5.2.4.2.5 Dynamic input.- Repeat paragraph 4.5.2.4.2.1 and 4.5.2.4.2.2 and apply to pin 61B a 10 cps $\pm$ 10 percent square wave varying from 0 $\pm$ 1 volt to minus 6 $\pm$ 2 volts as shown in Figure 7. The output wave form at pin 65A shall be as shown in Figure 6. The delay time measured from the time the input is at minus 3 volts (going toward minus 6 volts), to the time the output reaches plus 90 volts shall be less than 10 milliseconds. The delay time measured from the time the input is at minus 3 volts (going toward 0 volts) to the time the output reaches 90 volts shall be less than 10 milliseconds for compliance with 3.4.3.4.

4.5.2.4.2.6 Circuit output.- The neon driver No. 1 shall be subjected to the test of paragraph 4.5.2.4.2.5, when thermally stabilized at ambient temperatures of minus 40 degrees to plus 135 degrees Fahrenheit for compliance with 3.3.3.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:18Z
Check the source to verify that this is the current version before use.

4.5.2.5 Neon driver No. 11.

4.5.2.5.1 Input diodes.

4.5.2.5.1.1 Forward voltage drop.- Connect a minus 18 volt supply voltage to pin 5B. The voltage measured from the following pins when each is grounded one at a time, to pin 57B shall be less than +0.5 volt. Pins 58B and 59B to conform with 3.4.4.1.

4.5.2.5.1.2 Reverse leakage.- Remove the minus 18 volts from pin 5B and ground pin 5B. The voltage measured from pin 5B to pin 57B when a minus 18 volt is applied one at a time to pins 58B and 59B, shall be less than 2 volt to conform with 3.4.4.1. Remove all test leads upon completion of test.

4.5.2.5.2 Threshold diodes.

4.5.2.5.2.1 Forward voltage drop.- Connect a minus 3 volt supply voltage to pin 5B and ground pin 54B. The voltage measured from pins 54B to TP16 shall be less than 0.5 volt to conform with 3.4.4.2.

4.5.2.5.2.2 Reverse leakage.- Remove the minus 3 volts and ground from pins 5B and 54B. Connect a minus 18 volts to pin 54B and ground pin 5B. The voltage measured from pin 5B to TP16 shall be less than +0.5 volt to conform with 3.4.4.2. Remove all test leads upon completion of test.

4.5.2.5.3 Output transistor.

4.5.2.5.3.1 Load.- Connect to pin 52B, the load shown in Figure 5.

4.5.2.5.3.2 Power supplies.- Connect power to the appropriate pin as shown in Table V.

4.5.2.5.3.3 True input.- Connect a minus 6 volt supply voltage to pin 58B. The voltage measured from TP17 to ground shall be plus 112 $\pm$ 15 volt to conform with 3.4.4.3.

4.5.2.5.3.4 False input.- Remove the minus 6 volts from pin 58B and ground pin 58B. The voltage measured from the points listed in Column 1 to those in Column 2 shall be as listed in Column 3 of Table XI to determine compliance with 3.4.4.3 and 3.4.4.4.

TABLE XI

COLUMN 1, FROM PIN	COLUMN 2, TO PIN	COLUMN 3, VOLTAGE
57B	TP16	$<$ +1.0
TP17	23B	+23 $\pm$ 15
52B	23B	+75 $\pm$ 9

4.5.2.5.3.5 Dynamic input.- Repeat paragraph 4.5.2.5.3.1 and 4.5.2.5.3.2 and apply to pin 58B a 10 cps $\pm$ 10 percent square wave varying from 0 $\pm$ 1 volts to minus 6 $\pm$ 2 volts as shown in Figure 7. The output waveform at pin 52B shall be as shown in Figure 6. The delay measured from the time the input is at minus 3 volts (going toward minus 6 volts) to the time when the output

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:18Z
Check the source to verify that this is the current version before use.

reaches plus 90 volts, shall be less than 10 milliseconds. The delay measured from the time the input is at minus 3 volts (going toward 0 volts), to the time when the output reaches plus 90 volts, shall be less than 10 milliseconds to conform with 3.4.4.5.

4.5.2.5.3.6 Circuit output.- The neon driver No. II shall be subjected to the test of paragraph 4.5.2.5.3.5, when thermally stabilized at ambient temperatures of minus 40 degrees to plus 135 degrees Fahrenheit for compliance with 3.3.3.1.

4.5.2.5.4 Logic driver.- To determine compliance with 3.4.5, the following test conditions shall be applied.

4.5.2.5.4.1 Input gates.- Prior to energizing the module, an input gate shown in Figure 9 shall be connected to pin 3B.

4.5.2.5.4.2 Power supplies.- Connect power supplies to appropriate pins as shown in Table V.

4.5.2.5.4.3 True state static conditions, driver unloaded.- Connect a minus 6 volt supply voltage to the input of the input gate and measure voltage as follows for compliance with 3.4.5.1(a) through (c):

(a) The voltage measured from pin 2A to pin 4B shall be less than +0.3 volt D.C. to conform with 3.4.5.1(a).

(b) The voltage measured from pin 1B to pin 52A shall be less than +0.015 volt D.C. to conform with 3.4.5.1(b).

(c) The voltage measured from pin 1A to pin 23B shall be plus 0.85 $\pm$ 0.65 volt D.C. to conform with 3.4.5.1(c).

4.5.2.5.4.4 False state static condition, driver loaded.- For the remainder of the tests in 4.5.2.5.4, output load as shown in Figure 10, shall be connected to pin 1B. Remove the minus 6 volts from the input to the input gate and ground the input. The following measured voltage shall determine compliance with 3.4.5.2(a) through (c).

(a) The voltage measured from pin 1A to pin 4B shall be plus 6.3 $\pm$ .6 volts D.C. to conform with 3.4.5.2(a).

(b) The voltage measured from pin 23B to pin 1B shall be less than +0.3 volt D.C. to conform with 3.4.5.2(b).

(c) The voltage measured from pin 23B to pin 1A shall be less than +0.7 volt D.C. to conform with 3.4.5.2(c). Remove ground from the input to the input gate.

4.5.2.5.4.5 Dynamic condition, driver loaded.

4.5.2.5.4.5.1 Test signal.- The input test signal to be used for the dynamic tests of the driver, shall be an approximately symmetrical square wave (each half period shall be 2.00 $\pm$ 0.2 microseconds) varying from minus 0.2 $\pm$ 0.2 volts to minus 6 $\pm$1volts with 10 to 90 percent rise and fall time each less than 0.4 microseconds to conform with 3.4.5.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:18Z
Check the source to verify that this is the current version before use.

4.5.2.5.4.5.2 Apply the test signal of paragraph 4.5.2.5.4.5.1 to the input of the input gate and observe the output wave form at pin 1B. The waveform shall be a square wave swinging between the levels of minus 0.3 ± 0.3 volt and minus 6 ± 1.0 volt at the frequency of the input test signal to conform with 3.4.5.3.

4.5.2.5.4.5.3 Measurements of delay time between inputs and outputs wave forms.- Compare the input test signal applied to the input gate and the output waveform at pin 1B. The delay time measured from the time the square wave applied to the input reaches minus 1.5 volts (going towards minus 6 volts) until the output reaches 90 percent of its total excursion (going towards minus 6 volts) shall be less than 150 millimicroseconds. The delay time measured from the time the square wave applied to the input reaches minus 1.5 volts (going towards zero volts) until the output reaches 90 percent of its total excursion (going toward zero volts) shall be less than 150 millimicroseconds to conform with 3.4.5.4 (see Figure 11).

4.5.2.5.4.6 Circuit output.- The logic driver shall be subjected to the tests of paragraphs 4.5.2.5.4.5 when thermally stabilized at ambient temperatures of plus 135 degrees and minus 40 degrees Fahrenheit for compliance with 3.3.3.1.

4.5.2.5.5 Performance for matrix decoder back contact clamp, ground clamps, "And" gates, gates and diodes.- The following tests shall be conducted to determine compliance with 3.4.6.

4.5.2.5.5.1 Reverse leakage.- Disconnect all previously connected leads. Apply to pin 25B a minus 18 volt supply voltage. Ground the pins shown in column 1 and measure the voltage from those pins listed in column 2 to pin 25B, the voltage shall be as listed in column 3 of Table XII to determine compliance with the requirements of 3.4.6.1. Remove ground after each measurement.

TABLE XII

1 GROUND PIN	2 VOLTAGE FROM PIN	3 VOLTAGE
3A	7A	≤+0.4 V.D.C.
5A	7A	≤+0.4 V.D.C.
6A	7A	≤+0.4 V.D.C.
8A	7A	≤+0.4 V.D.C.
12A	13A	≤+0.4 V.D.C.
14A	13A	≤+0.4 V.D.C.
15A	13A	≤+0.4 V.D.C.
18A	13A	≤+0.4 V.D.C.
22A	20A	≤+0.4 V.D.C.
23A	20A	≤+0.4 V.D.C.
25A	20A	≤+0.4 V.D.C.
26A	20A	≤+0.4 V.D.C.
23B	7A, 13A, 20A, 4A 11A, 19A	≤+0.4 V.D.C.
	32A	≤+0.4 V.D.C.
10A	4A	≤+0.4 V.D.C.
17A	11A	≤+0.4 V.D.C.
24A	19A	≤+0.4 V.D.C.

Source: https://assist.dla.mil -- Downloaded: 2025-10-23 10:07 EDT
Check the source to verify that this is the current version before use.

4.5.2.5.5.1.1 Remove the minus 18 volts from pin 25B and connect the plus 15 volt supply voltage to pin 5B. Ground the pins listed in column 1 and measure the voltage from pin 5B to the pins listed in Column 2. The voltage shall be the value listed in Column 3 of Table XIII to determine compliance with the requirements of 3.4.6.1 and 3.4.6.2.

TABLE XIII

1 GROUND PIN	2 VOLTAGE TO PIN	3 VOLTAGE
46A	38A, 48A	≤+0.5
44A	38A, 48A	≤+0.5
41A	38A	≤+0.5
40A	38A	≤+0.5
52A	38A, 48A, 56A, 60A	≤+0.15
50A	48A	≤+0.5
51A	48A	≤+0.5
58A	56A, 60A	≤+0.5
59A	56A, 60A	≤+0.5
57A	56A	≤+0.5
61A	56A	≤+0.5
62A	60A	≤+0.5
63A	60A	≤+0.5
55A	54A	≤+0.5

4.5.2.5.5.2 Forward voltage drop.- Remove the plus 15 volts from pin 5B and connect a plus 15 volt supply voltage to pin 25B. Ground the pins in column 1 and measure the voltage from those pins listed in column 2 to ground and the voltage shall be as listed in column 3 of Table XIV to determine compliance with the requirements of 3.4.6.1.

TABLE XIV

1 GROUND PIN	2 VOLTAGE FROM PIN	3 VOLTAGE
Pin 3A	Pin 7A	≤+1.0
5A	7A	≤+1.0
6A	7A	≤+1.0
8A	7A	≤+1.0
23B	7A, 13A, 20A	≤+1.0
23B	4A, 11A, 19A, 32A	≤+1.0
12A	13A	≤+1.0
14A	13A	≤+1.0
15A	13A	≤+1.0
18A	13A	≤+1.0
22A	20A	≤+1.0
23A	20A	≤+1.0
25A	20A	≤+1.0
26A	20A	≤+1.0
10A	4A	≤+1.0
17A	11A	≤+1.0
24A	19A	≤+1.0

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:18Z
Check the source to verify that this is the current version before use.

4.5.2.5.5.2.1 Remove the plus 15 volts from pin 25B and connect a minus 18 volt supply voltage to pin 5B. Ground the pin listed in column 1 and measure the voltage from ground to those pins listed in column 2. The voltage shall be as listed in column 3 of Table XV to determine compliance with the requirements of 3.4.6.1 and 3.4.6.2.

TABLE XV

1 GROUND PIN	2 VOLTAGE AT PIN	3 VOLTAGE
46A	38A, 48A	≤+1.0
44A	38A, 48A	≤+1.0
41A	38A	≤+1.0
40A	38A	≤+1.0
52A	38A, 48A 56A, 60A	≤+1.0
50A	48A	≤+1.0
51A	48A	≤+1.0
58A	56A, 60A	≤+1.0
59A	56A, 60A	≤+1.0
57A	56A	≤+1.0
61A	56A	≤+1.0
62A	60A	≤+1.0
63A	60A	≤+1.0
55A	54A	≤+1.0

4.5.2.5.5.3 Spare diode test.- Remove the minus 18 volts from pin 5B. Connect a 2.2K ohms, ± 5 percent, 1/2 watt, resistor from pin 34A to ground. Apply a minus 18 volt supply voltage to pin 31A. The voltage measured from ground to pin 34A shall be less than plus 0.2 volt D.C. Remove the minus 18 volts from pin 31A and apply a plus 15 volt supply voltage to pin 31A. The voltage measured from pin 31A to pin 34A shall be less than plus 1.0 volt dc. Remove all connections to pins 31A and 34A.

4.5.2.5.5.3.1 Repeat paragraph 4.5.2.5.5.3 except pin 28A was 31A and pin 29A was pin 34A.

4.5.3 Wiring.- Prior to the performance testing, the network shall be visually inspected to insure that all components are physically present and connected in conformance with the drawing specified in 3.5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.- Preservation, packaging and packing shall be in accordance with Master Packaging Data Sheet M14002.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:18Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use.- The Network A assembly is made up of miscellaneous circuits; the use of each circuit as described:

(a) Output driver.- The output driver is used to supply information to circuits external to the computer by converting logical levels to field data levels.

(b) Input driver.- The input driver is used to convert information signals from circuits external to the computer to computer logical levels.

(c) Neon driver No. 1.- The neon driver No. 1 is used to turn on and off neon indicators on the face of the control panel to indicate malfunction of the computer.

(d) Neon driver No. 11.- The neon driver No. II is used to turn on and off neon indicators on the face of the control panel to indicate malfunctions of the computer. This differs from the neon driver No. I by the type of signal to its input.

(e) Logic driver.- The logic driver is capable of furnishing a power amplification of 26 to the computer logic signals.

(f) Matrix decoder, back contact clamps, "And" gates, ground clamps, and gate.- These circuits are connected to various points in the computer to mechanize the logical processes which the computer performs.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of applicable level of preservation, packaging and packing
(c) Applicable stock number
(d) Applicable packaging data sheets (see 5.1)
(e) A block of serial numbers to identify each network board

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1220-A202Y

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:18Z
Check the source to verify that this is the current version before use.